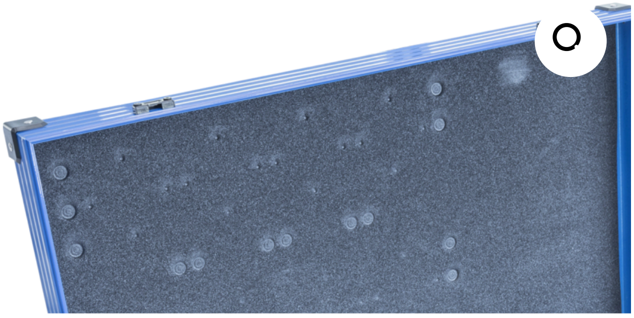

RE18 – Advanced Smart Grid Technology

The RE18 Advanced Smart Grid Technology system provides students with a self-contained modular system, allowing an in-depth understanding of the complex interactions between renewable energies, energy storage and consumers on a laboratory scale.

The system allows the construction of a controllable Smart Grid on a laboratory scale. Providing multiple pre-set scenarios but additionally allows for bespoke or user-created setups.

PRODUCT CODE: **RE18**

Categories: Educational, RE Series - Renewable Energy, RE18 Smart Grid Technology

Tags: Consumer Load , Grid Stability , Laboratory Scale , Lithium-Iron-Phosphate Batteries , Photovoltaic , Power Fluctuations , Power Plant , PV , Radial Distribution , RE18 , Renewable Energy , Smart Grid Technology , Voltage Behaviour , Wind Power

DESCRIPTION

The RE18 Advanced Smart Grid Technology system provides students with a self-contained modular system, allowing an in-depth understanding of the complex interactions between renewable energies, energy storage and consumers on a laboratory scale.

The system allows the construction of a controllable Smart Grid on a laboratory scale. Providing multiple pre-set scenarios but additionally allows for bespoke or user-created setups.

Contained in the system are multiple components for renewable energies such as wind and photovoltaics as well as energy stores such as lithium-iron-phosphate batteries / fuel cells that allow a large variety of fundamental experiments to be conducted in conjunction with smart grid experiments.

TECHNICAL SPECIFICATIONS

- 2 x Base unit Professional
- 2 x Smart meter
- 1 x Wind turbine module Pro

- 1 x Solar module 5.33V, 370mA
- 1 x Wind machine
- 1 x Motor module Pro
- 1 x Base for solar panel
- 2 x Power module
- 1 x Wind rotor set
- 2 x Light bulb module Pro
- 1 x Capacitor module Pro
- 1 x AV-module
- 1 x Battery module holder 1xAAA Pro
- 1 x LiFePo-battery AAA
- 1 x Fuel cell holder Pro
- 1 x MPP Tracker Pro
- 2 x Grid module Pro
- 1 x Diode module Pro
- 1 x Potentiometer module 110 Ohm Pro
- 1 x Bulb 120W, 12°
- 1 x Lamp housing
- 6 x Safety short-circuit plug, with mid socket
- 1 x Aluminium case
- 5 x Safety test lead, 25cm, red
- 4 x Safety test lead, 25cm, black
- 4 x Safety test lead, 50cm, red
- 4 x Safety test lead, 50cm, black
- 1 x Propeller
- 1 x Reversible fuel cell
- 1 x Azimuth angle scale

FEATURES & BENEFITS

- Setup of a complete smart grid on a laboratory scale
- Investigating the influence of renewable energies on grid stability
- Additional fundamental experiments on wind, photovoltaics, fuel cells and energy storage
- Laboratory scale
- Modular design
- Supplied in a self contained aluminium case
- Includes in-depth manual and predefined experiments

Demonstration Capabilities

Smart grid experiments:

- Daily power fluctuations of a photovoltaic (PV) power plant
- Daily power fluctuations of a wind power plant
- Energy supply of a building by conventional power plants
- Energy supply of a building by conventional and PV power plants
- Energy supply of a building by conventional and PV power plants with storage
- Voltage behaviour and grid stability in a radial distribution system
- Grid stability with PV power plants

- Grid stability with PV power plants depending on consumer load
- Grid stability with PV power plants depending on wire length
- Grid stability with PV power plants and smart transformer stations
- Grid stability with PV power plants and storages
- Grid integration of E-Mobility
- Conductor rope management

Fundamental experiments:

Photovoltaics

- IV-Characteristics of solar panels
- IV-Characteristics depending on illumination
- IV-Characteristics depending on temperature
- MPP-Tracking

Wind energy

- Turbine power dependent on blade shape and pitch angle
- Turbine power dependent on number of blades
- Turbine power dependent on wind direction

Fuel cell and electrolyzer

- Functionality of an electrolyzer
- IV-characteristics of an electrolyzer
- Functionality of a fuel cell
- IV-characteristics of a fuel cell

Storage technologies

- Charge and discharge characteristics of a capacitor
- Functionality and charge procedure of a LiFePo battery
- Operation of fuel cells and electrolysers

Related curriculums and Video

- Renewable Energies
- Electrical Engineering

Other products in the series

- **RE10:** Advanced Photovoltaic Energy
- **RE12:** Advanced Wind Energy
- **RE14:** Advanced Fuel Cell Technology
- **RE16:** Advanced Thermal Energy
- **RE24:** Advanced Battery Technology

Operational conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz

- Level and stable work surface

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 10Kg

Dimensions

- **Length:** 0.640m
- **Width:** 0.165m
- **Height:** 0.370m

Order Codes

RE18: Advanced Smart Grid Technology

YOU MAY ALSO LIKE…

RE12 – ADVANCED WIND ENERGY

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RE24 – ADVANCED BATTERY TECHNOLOGY

Read more

RE14 – ADVANCED FUEL CELL TECHNOLOGY

Read more

RE16 – ADVANCED THERMAL ENERGY

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RE 10 – ADVANCED PHOTOVOLTAIC ENERGY

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